Linux Fundamentals: The Night Shift — Instructor Guide

Every challenge, answer, and accepted variant · generated 2026-08-25 from the live pack data

Quick reference
Set your own site URL and class password before handing this out. The admin handle is whatever you put in `ADMIN_HANDLES` — `instructor`, say; there is no default, and the setting ships empty.
Handles: 3–20 characters, letters/numbers/underscore/hyphen, **claimed once only**.
46 challenges · 1155 points total · finds look like `FIND{ABCD2345EFGH}` and are **different for every student**.

How the difficulty fades — Act I briefs show the whole command, and the free hint repeats it. From Act II the brief names the tool but not its flags or arguments, and the exact line moves into a hint that costs XP. By the last act the brief states only the objective. A student who never buys a hint has genuinely recalled every command. Hint spend is visible in the instructor dashboard, so it reads as a difficulty signal rather than as cheating.

If a student says "it's not working"

Symptom	Cause & what to say
"My flag is rejected"	Flags are per-student — a classmate's flag never validates. Tell them to **click the green flag in the terminal** to copy it exactly, rather than retyping.
"Handle already claimed"	Someone took it, or they registered before and cleared their browser. Pick a new handle. (To free one, delete that player in Netlify Blobs.)
"I ran the command, nothing happened"	Check the blue path in their prompt — they may be in a different directory. Most challenges accept short or full paths, but the file must exist from where they stand. `cd ~` returns home.
"It says command not found"	If it's a real command we don't simulate (`top`, `nano`), the game says so explicitly. Genuine typos say "command not found".
"I don't know which command to use"	Intended from Act II on. Point them at the free hint first, then the Reference tab. The costed hint is the safety net, not a failure.
"The act is locked"	Each act opens after solving all but one of the previous act's challenges.
"Nothing printed"	Correct for `cd` and redirects. The game prints "(no output — in a shell, silence usually means success)".
Lost all progress	Progress is tied to the browser token. Same browser = resumes automatically. Different machine = new handle needed.

Act structure & unlocking

Act	Teaches	Challenges	Points	Unlock
Act I: Opening the Dome	Learn the building before dark: paths, listings, files, and wildcards	11	185	Always open
Act II: Reading the Night Log	Filter, count, sort and pipe what the observatory recorded	13	340	Solve 9 of 11 in the prior act

Act III: Keys to the Dome	Create, copy, move, remove, and decide who may read what	12	310	Solve 9 of 13 in the prior act
Act IV: Handover at Dawn	Find, edit, test and chain — everything the day crew will not do for you	10	320	Solve 9 of 12 in the prior act

Students may skip **one** challenge per act and still advance.

Act I: Opening the Dome

Learn the building before dark: paths, listings, files, and wildcards

Where Are You Standing?

10 pts · l1-pwd · OK

Task: First thing on shift: put your position on the record. Run `pwd` to print the full path of the directory you are standing in.

Answer: `pwd`

Completion: Auto-completes on: allOf

Hint 1 (free): Type `pwd` and press Enter.

What Is In Here?

10 pts · l1-ls · OK

Task: Look around before you touch anything. Run `ls` to list the files and folders in the directory you are standing in.

Answer: `ls`

Completion: Auto-completes on: allOf

Hint 1 (free): Type `ls`.

Also accepted: `ls .`, `ls ./`

Reveal Hidden Dotfiles

15 pts · l1-ls-la · OK

Task: On Linux a filename that begins with a dot is hidden from a plain listing, and the observer before you left something in your home directory with exactly that kind of name. Use `ls -la` to list everything, hidden files included.

Answer: `ls -la` **then** `ls -al`

Completion: Auto-completes on: allOf

Hint 1 (free): Run `ls -la` or `ls -al`.

Also accepted: `ls -al`, `ls -a -l`, `ls -a`

Changing Directories

15 pts · l1-cd · OK

Task: Tonight's paperwork lives in `Documents`. Run `cd Documents` to move into it.

Answer: `cd Documents`

Completion: Auto-completes on: cwdIs

Hint 1 (free): Type `cd Documents`.

Also accepted: `cd ./Documents`, `cd /home/student/Documents`, `cd ~/Documents`

Climbing Up (..)

15 pts · l1-cd-parent · OK

Task: The symbol `..` always means the parent directory. From `Documents`, run `cd ..` to get back to your home directory.

Answer: `cd ..`

Completion: Auto-completes on: cwdIs

Hint 1 (free): Type `cd ..`.

Also accepted: `cd ../`, `cd ~`, `cd /home/student`, `cd`

Read the Night Notes

15 pts · l1-cat · OK

Task: Print the whole of `Documents/notes.txt` with `cat` and find out what tonight's target is.

Answer: `cat Documents/notes.txt`

Completion: Auto-completes on: outputEquals

Hint 1 (free): Run `cat Documents/notes.txt`.

Also accepted: `cat ./Documents/notes.txt`, `cat /home/student/Documents/notes.txt`

First Few Lines (head)

15 pts · l1-head · OK

Task: `Documents/data.csv` is the duty roster. Show only its first 3 lines with `head` — the column names, and the first two people on it.

Answer: `head -n 3 Documents/data.csv`

Completion: Auto-completes on: outputEquals

Hint 1 (free): Run `head -n 3 Documents/data.csv`.

Also accepted: `head -3 Documents/data.csv`

Last Few Lines (tail)

15 pts · l1-tail · OK

Task: Show only the last 2 lines of `Documents/data.csv` with `tail`.

Answer: `tail -n 2 Documents/data.csv`

Completion: Auto-completes on: outputEquals

Hint 1 (free): Run `tail -n 2 Documents/data.csv`.

Also accepted: `tail -2 Documents/data.csv`

Wildcard Globbing (*)

20 pts · l1-glob · OK

Task: The observatory website lives in `projects/web`. List every `.js` file in there with a single `*` pattern instead of typing filenames: `ls projects/web/*.js`.

Answer: `ls projects/web/*.js`

Completion: Auto-completes on: allOf

Hint 1 (free): Run `ls projects/web/*.js`.

Also accepted: `ls ./projects/web/*.js`

Multiple Wildcards

20 pts · l1-glob-doc · OK

Task: List every `.txt` file in `Documents` with one wildcard pattern.

Answer: `ls Documents/*.txt`

Completion: Auto-completes on: allOf

Hint 1 (free): Run `ls Documents/*.txt`.

Also accepted: `ls ./Documents/*.txt`

BOSS: The Handover Note

35 pts · l1-boss · OK

Task: The observer before you left a note in your home directory, and a plain listing will not show it. Find it, read it, and submit the dome key written inside — it was generated for you alone, so nobody else can hand it to you.

Answer: `cat .handover`

Completion: Submits a flag · flag is inside `/home/student/.handover`

Hint 1 (free): A hidden filename starts with a dot, and one `ls` find lists those too. Once you can see the name, print the file.

Hint 2 (−5 XP): The file is `.handover`. Read it with `cat`, then paste the FIND{...} value into the box.

Also accepted: `cat /home/student/.handover`, `cat ~/.handover`, `ls -a && cat .handover`

Act II: Reading the Night Log

Filter, count, sort and pipe what the observatory recorded

Search Patterns with grep

20 pts · l2-grep · OK

Task: Pull every line out of `Documents/data.csv` that mentions the word `active`. `grep` is the tool that prints the lines matching a pattern.

Answer: `grep active Documents/data.csv`

Completion: Auto-completes on: outputEquals

Hint 1 (free): `grep` wants the pattern first and the file second.

Hint 2 (−10 XP): Run `grep active Documents/data.csv`.

Also accepted: `grep "active" Documents/data.csv`, `cat Documents/data.csv | grep active`

Case-Insensitive Grep (-i)

20 pts · l2-grep-i · OK

Task: The dome controller writes `logs/app_errors.log`, and it spells trouble three different ways: ERROR, Error and error. Find all three of them in one search.

Answer: `grep -i error logs/app_errors.log`

Completion: Auto-completes on: allOf

Hint 1 (free): One short `grep` find makes the match ignore letter case.

Hint 2 (−10 XP): Run `grep -i error logs/app_errors.log`.

Also accepted: `grep -i "error" logs/app_errors.log`, `grep -i ERROR logs/app_errors.log`

Inverting Matches (-v)

25 pts · l2-grep-v · OK

Task: `logs/app_errors.log` is mostly INFO noise. Show every line EXCEPT the ones containing INFO.

Answer: `grep -v INFO logs/app_errors.log`

Completion: Auto-completes on: allOf

Hint 1 (free): One `grep` find inverts the match, so it prints the lines that do NOT match.

Hint 2 (−10 XP): Run `grep -v INFO logs/app_errors.log`.

Also accepted: `grep -v "INFO" logs/app_errors.log`

Line Counting (wc -l)

20 pts · l2-wc-l · OK

Task: How many requests did the observatory website serve last night? Count the lines in `Documents/server_access.log`.

Answer: `wc -l Documents/server_access.log`

Completion: Auto-completes on: allOf

Hint 1 (free): `wc` counts lines, words and bytes. One find limits it to lines.

Hint 2 (−10 XP): Run `wc -l Documents/server_access.log`.

Also accepted: `wc -l ./Documents/server_access.log`, `cat Documents/server_access.log | wc -l`

Sorting Numbers (sort -n)

20 pts · l2-sort · OK

Task: `Documents/dome_temps.log` is one dome temperature reading per line, in the order the controller logged them. Put them in order, coldest first. `sort` reads text by default, and a two-digit reading can land in the wrong place unless it is told the column holds numbers.

Answer: `sort -n Documents/dome_temps.log`

Completion: Auto-completes on: allOf

Hint 1 (free): `sort` compares lines as text by default. One find tells it to compare them as numbers instead.

Hint 2 (−10 XP): Run `sort -n Documents/dome_temps.log`.

Also accepted: `sort -n ./Documents/dome_temps.log`

Sorting by Field (sort -t / -k)

25 pts · l2-sort-field · OK

Task: `Documents/data.csv` is comma separated, and column 4 is the hours each person logged. Put the roster in order by that column, smallest number first. You already know the find that keeps it numeric; `sort` also needs to be told the separator and which column to use.

Answer: `sort -t, -k4 -n Documents/data.csv`

Completion: Auto-completes on: allOf

Hint 1 (free): Two more finds: one sets the field separator, one picks the field number. Comma is the separator here, and hours is the fourth field.

Hint 2 (−10 XP): Run `sort -t, -k4 -n Documents/data.csv`.

Also accepted: `sort -t, -n -k4 Documents/data.csv`, `sort -n -t, -k4 Documents/data.csv`

Field Extraction (cut)

25 pts · l2-cut · OK

Task: From `Documents/data.csv`, print only column 2 (the name) and column 4 (the hours). `cut` slices columns out of delimited text.

Answer: `cut -d, -f2,4 Documents/data.csv`

Completion: Auto-completes on: allOf

Hint 1 (free): One find sets the delimiter, another picks the fields. Separate the two field numbers with a comma and no space.

Hint 2 (−10 XP): Run `cut -d, -f2,4 Documents/data.csv`.

Also accepted: `cut -d"," -f2,4 Documents/data.csv`, `cut -d',' -f2,4 Documents/data.csv`

The Pipeline (grep | wc -l)

30 pts · l2-pipe-wc · OK

Task: How many requests in `Documents/server_access.log` came back with status 401? Do it in one line: find the matching lines, then feed them straight into a counter.

Answer: `grep 401 Documents/server_access.log | wc -l`

Completion: Auto-completes on: allOf

Hint 1 (free): The `|` operator sends one command's output into the next one as its input.

Hint 2 (−10 XP): Run `grep 401 Documents/server_access.log | wc -l`.

Also accepted: `grep "401" Documents/server_access.log | wc -l`, `grep -c 401 Documents/server_access.log`, `cat Documents/server_access.log | grep 401 | wc -l`

Output Redirection (>)

30 pts · l2-redirect-out · OK

Task: Save every error line from `logs/app_errors.log` — all three spellings — into a new file at `/tmp/critical_errors.txt`. Nothing should appear on the screen.

Answer: `grep -i error logs/app_errors.log > /tmp/critical_errors.txt`

Completion: Auto-completes on: allOf

Hint 1 (free): `>` sends a command's output into a file instead of the screen.

Hint 2 (−10 XP): Run `grep -i error logs/app_errors.log > /tmp/critical_errors.txt`.

Also accepted: `grep -i "error" logs/app_errors.log > /tmp/critical_errors.txt`

Append Redirection (>>)

30 pts · l2-redirect-append · OK

Task: Add the line `Audit log verified` to the END of `/tmp/audit_trail.log` without erasing anything already in it.

Answer: `echo "Audit log verified" >> /tmp/audit_trail.log`

Completion: Auto-completes on: allOf

Hint 1 (free): `>` overwrites. Doubling that same operator appends instead. `echo` produces the text to append.

Hint 2 (−10 XP): Run `echo "Audit log verified" >> /tmp/audit_trail.log`.

Also accepted: `echo Audit log verified >> /tmp/audit_trail.log`

Split Output (tee)

35 pts · l2-tee · OK

Task: Show every `Weather` line from `Documents/data.csv` on screen AND save those same lines to `/tmp/weather_team.txt`, in one command.

Answer: `grep Weather Documents/data.csv | tee /tmp/weather_team.txt`

Completion: Auto-completes on: allOf

Hint 1 (free): `tee` writes what it receives to a file and passes it on to the screen at the same time.

Hint 2 (−10 XP): Run `grep Weather Documents/data.csv | tee /tmp/weather_team.txt`.

Also accepted: `grep "Weather" Documents/data.csv | tee /tmp/weather_team.txt`

Collapsing Repeats (uniq)

20 pts · l2-uniq · OK

Task: The dome controller writes its status to `Documents/dome_status.log` every time it polls, whether or not anything changed. Show the run of readings with the repeats collapsed, so only the actual status changes are left.

Answer: `uniq Documents/dome_status.log`

Completion: Auto-completes on: allOf

Hint 1 (free): `uniq` drops a line that is an exact repeat of the one directly above it. Repeats that are not next to each other are left alone.

Hint 2 (−10 XP): Run `uniq Documents/dome_status.log`.

Also accepted: `cat Documents/dome_status.log | uniq`

BOSS: How Many Machines Called?

40 pts · l2-boss · OK

Task: `Documents/server_access.log` holds one request per line, and the first field of each line is the address that made it. Several addresses appear more than once. Print how many DIFFERENT addresses appear in the file — one number, nothing else.

Answer: `cut -d' ' -f1 Documents/server_access.log | sort | uniq | wc -l`

Completion: Auto-completes on: allOf

Hint 1 (free): Four jobs on one line: take the first field of every line, put those values in order, collapse the repeats, then count what is left.

Hint 2 (−15 XP): Run `cut -d' ' -f1 Documents/server_access.log | sort | uniq | wc -l`.

Also accepted: `cut -d" " -f1 Documents/server_access.log | sort -u | wc -l`, `awk '{print $1}' Documents/server_access.log | sort | uniq | wc -l`

Act III: Keys to the Dome

Create, copy, move, remove, and decide who may read what

Create Empty Files (touch)

20 pts · l3-touch · OK

Task: Open a scratch file for the night: create an empty file at `/tmp/new_file.txt`.

Answer: `touch /tmp/new_file.txt`

Completion: Auto-completes on: fileExists

Hint 1 (free): `touch` creates an empty file, or updates the timestamp of one that already exists.

Hint 2 (−10 XP): Run `touch /tmp/new_file.txt`.

Make Directories (mkdir -p)

20 pts · l3-mkdir · OK

Task: Make a new directory `projects/api`. Use the find that also creates any missing parent, so the command works whether or not `projects` is already there.

Answer: `mkdir -p projects/api`

Completion: Auto-completes on: dirExists

Hint 1 (free): `mkdir` makes directories. One find creates the missing parents and stays quiet if the directory already exists.

Hint 2 (−10 XP): Run `mkdir -p projects/api`.

Also accepted: `mkdir projects/api`, `mkdir -p ./projects/api`

Copy Files (cp)

20 pts · l3-cp · OK

Task: Back up tonight's notes: copy `Documents/notes.txt` to `/tmp/notes_backup.txt`.

Answer: `cp Documents/notes.txt /tmp/notes_backup.txt`

Completion: Auto-completes on: allOf

Hint 1 (free): `cp` takes the source path first and the destination second.

Hint 2 (−10 XP): Run `cp Documents/notes.txt /tmp/notes_backup.txt`.

Also accepted: `cp ./Documents/notes.txt /tmp/notes_backup.txt`

Recursive Copy (cp -r)

25 pts · l3-cp-r · OK

Task: The day crew wants the whole observatory website exactly as it stands tonight — every file in it, not only the ones you happened to touch.

Answer: `cp -r projects/web /tmp/web_backup`

Completion: Auto-completes on: allOf

Hint 1 (free): `cp` refuses a directory until you add the recursive find.

Hint 2 (−10 XP): Run `cp -r projects/web /tmp/web_backup`.

Also accepted: `cp -R projects/web /tmp/web_backup`

Move & Rename (mv)

20 pts · l3-mv · OK

Task: The deploy script is about to be pointed at the production dome. Its name should say so before somebody runs it by accident at three in the morning.

Answer: `mv projects/scripts/deploy.py projects/scripts/deploy_prod.py`

Completion: Auto-completes on: fileExists

Hint 1 (free): `mv` both moves and renames: old path first, new path second. Renaming is just moving to a new name in the same place.

Hint 2 (−10 XP): Run `mv projects/scripts/deploy.py projects/scripts/deploy_prod.py`.

Also accepted: `mv ./projects/scripts/deploy.py ./projects/scripts/deploy_prod.py`

Remove Files (rm)

20 pts · l3-rm · OK

Task: `logs/debug.log` is noise and the day crew wants it gone. Delete it — and on the same line list `logs` again, so you can see for yourself that it went.

Answer: `rm logs/debug.log && ls logs`

Completion: Auto-completes on: allOf

Hint 1 (free): `rm` removes a file. There is no undo and no recycle bin, so checking afterwards is a habit worth having.

Hint 2 (−10 XP): Run `rm logs/debug.log && ls logs`.

Also accepted: `rm -f logs/debug.log && ls logs`

Make Executable (chmod +x)

25 pts · l3-chmod-symbolic · OK

Task: `projects/scripts/deploy.py` cannot be run yet. Give it execute permission.

Answer: `chmod +x projects/scripts/deploy.py`

Completion: Auto-completes on: anyOf

Hint 1 (free): `chmod` also takes symbolic modes: a `+` or a `-`, then a letter for the permission.

Hint 2 (−10 XP): Run `chmod +x projects/scripts/deploy.py`.

Also accepted: `chmod u+x projects/scripts/deploy.py`, `chmod a+x projects/scripts/deploy.py`, `chmod 755 projects/scripts/deploy.py`

Permissions (chmod octal)

30 pts · l3-chmod-numeric · OK

Task: Lock `Documents/notes.txt` down so the owner can read and write it and nobody else can do anything at all. Set the octal mode.

Answer: `chmod 600 Documents/notes.txt`

Completion: Auto-completes on: fileHasMode

Hint 1 (free): Read is 4, write is 2, execute is 1. Add them up per column: owner, then group, then everybody else.

Hint 2 (−10 XP): Run `chmod 600 Documents/notes.txt`.

Also accepted: `chmod 600 ./Documents/notes.txt`

Inspect Inodes & Timestamps (stat)

20 pts · l3-stat · OK

Task: Show the size, permissions, inode number and timestamps of `Documents/notes.txt`. A plain `ls -l` does not show all of that.

Answer: `stat Documents/notes.txt`

Completion: Auto-completes on: allOf

Hint 1 (free): One command prints a file's whole metadata block in one go.

Hint 2 (−10 XP): Run `stat Documents/notes.txt`.

Also accepted: `stat ./Documents/notes.txt`

Superuser Elevation (sudo)

35 pts · l3-sudo-shadow · OK

Task: `/etc/shadow` holds the account password hashes and only root may read it, so `cat /etc/shadow` is refused. Read it anyway.

Answer: `sudo cat /etc/shadow`

Completion: Auto-completes on: allOf

Hint 1 (free): One command runs a single other command as a different user, normally root.

Hint 2 (−10 XP): Run `sudo cat /etc/shadow`.

Also accepted: `sudo head -n 2 /etc/shadow`

Instructor note: The brief deliberately shows the command that FAILS. Elevation is the lesson.

Conditional AND (&&)

30 pts · l4-list-and · OK

Task: On one line: create the directory `/tmp/test_dir`, then create the file `/tmp/test_dir/file.txt` inside it — but only attempt the file if the directory step succeeded.

Answer: `mkdir -p /tmp/test_dir && touch /tmp/test_dir/file.txt`

Completion: Auto-completes on: allOf

Hint 1 (free): One operator runs the second command only when the first one exits successfully.

Hint 2 (−15 XP): Run `mkdir -p /tmp/test_dir && touch /tmp/test_dir/file.txt`.

Also accepted: `mkdir /tmp/test_dir && touch /tmp/test_dir/file.txt`

BOSS: Seal the Night's Records

45 pts · l3-boss · OK

Task: Before you hand over, put a copy of `logs/auth.log` into a new directory `/tmp/handover`, lock that copy so only its owner may read and write it, and list the directory in long form so the permissions are visible. One line, with the steps joined so that each runs only if the one before it worked.

Answer: `mkdir -p /tmp/handover && cp logs/auth.log /tmp/handover/ && chmod 600 /tmp/handover/auth.log && ls -l /tmp/handover`

Completion: Auto-completes on: allOf

Hint 1 (free): Four jobs on one line, chained with `&&`: make the directory, copy the file into it, change the copy's mode, then list the directory in long form.

Hint 2 (−15 XP): Run `mkdir -p /tmp/handover && cp logs/auth.log /tmp/handover/ && chmod 600 /tmp/handover/auth.log && ls -l /tmp/handover`.

Also accepted: `mkdir -p /tmp/handover && cp logs/auth.log /tmp/handover/auth.log && chmod 600 /tmp/handover/auth.log && ls -l /tmp/handover`

Act IV: Handover at Dawn

Find, edit, test and chain — everything the day crew will not do for you

Finding Files by Name

30 pts · l4-find-name · OK

Task: Somewhere below your home directory there is a shell script. List every file whose name ends in `.sh`, at any depth under the directory you are in.

Answer: `find . -name "*.sh"`

Completion: Auto-completes on: allOf

Hint 1 (free): One command walks an entire directory tree and can filter on the filename. Quote the pattern so the shell does not try to expand it first.

Hint 2 (−15 XP): Run `find . -name "*.sh"`.

Also accepted: `find . -name '*.sh'`, `find /home/student -name "*.sh"`, `find . -type f -name "*.sh"`

Filtering by Type

30 pts · l4-find-type · OK

Task: List every regular file underneath `projects` — the files themselves, not the directories that hold them.

Answer: `find projects -type f`

Completion: Auto-completes on: allOf

Hint 1 (free): `find` can filter by entry type. The letter for a regular file is `f`.

Hint 2 (−15 XP): Run `find projects -type f`.

Also accepted: `find ./projects -type f`, `find /home/student/projects -type f`

Stream Editing (sed)

35 pts · l4-sed-replace · OK

Task: Print `Documents/data.csv` with every occurrence of `Optics` replaced by `Astrometry`. The file on disk must not change.

Answer: `sed "s/Optics/Astrometry/g" Documents/data.csv`

Completion: Auto-completes on: allOf

Hint 1 (free): `sed` edits a stream as it goes past. Its substitute script has the form s/old/new/g.

Hint 2 (−15 XP): Run `sed "s/Optics/Astrometry/g" Documents/data.csv`.

Also accepted: `sed 's/Optics/Astrometry/g' Documents/data.csv`

Pattern Scanning (awk)

35 pts · l4-awk-column · OK

Task: `/etc/passwd` is colon separated. Print only the username (field 1) and the home directory (field 6) for every account on this machine.

Answer: `awk -F: '{print $1, $6}' /etc/passwd`

Completion: Auto-completes on: allOf

Hint 1 (free): `awk` splits each line into numbered fields. One find sets the separator, and `$1` refers to the first field.

Hint 2 (−15 XP): Run `awk -F: '{print $1, $6}' /etc/passwd`.

Variable Expansion & Quotes

25 pts · l4-var-quotes · OK

Task: Print the value your `HOME` variable holds. Quote it the way that still lets the shell expand it.

Answer: `echo "$HOME"`

Completion: Auto-completes on: allOf

Hint 1 (free): Double quotes still allow variable expansion. Single quotes suppress it and print the name literally.

Hint 2 (−15 XP): Run `echo "$HOME"`.

Also accepted: `echo $HOME`

Exit Codes ($?)

30 pts · l4-exit-status · OK

Task: Every program returns an exit code when it finishes: 0 means success, anything else means failure. Search `/etc/passwd` for `root`, then print the exit code that search returned — both on one line.

Answer: `grep "root" /etc/passwd; echo $?`

Completion: Auto-completes on: allOf

Hint 1 (free): `$?` holds the exit code of the command that just finished, and `;` separates two commands on one line.

Hint 2 (−15 XP): Run `grep "root" /etc/passwd; echo $?`.

Also accepted: `grep root /etc/passwd; echo $?`

Conditional OR (||)

30 pts · l4-list-or · OK

Task: On one line: try to read `/non_existent_file`, and if that attempt fails, print the message `File was missing`.

Answer: `cat /non_existent_file || echo "File was missing"`

Completion: Auto-completes on: allOf

Hint 1 (free): One operator runs the second command only when the first one fails.

Hint 2 (−15 XP): Run `cat /non_existent_file || echo "File was missing"`.

Also accepted: `cat /non_existent_file || echo 'File was missing'`

Compare Files (diff)

30 pts · l4-diff · OK

Task: Compare `projects/web/index.html` against `projects/web/style.css` line by line, in the unified format that patch tools read.

Answer: `diff -u projects/web/index.html projects/web/style.css`

Completion: Auto-completes on: allOf

Hint 1 (free): `diff` compares two files. One find selects the unified output format.

Hint 2 (−15 XP): Run `diff -u projects/web/index.html projects/web/style.css`.

Also accepted: `diff -u ./projects/web/index.html ./projects/web/style.css`

Shell Mastery (history) 25 pts · l4-history · OK

Task: Your handover note has to say what you actually did tonight, and the shell has been keeping that list for you the whole shift.

Answer: `history`

Completion: Auto-completes on: allOf

Hint 1 (free): The shell keeps a numbered list of everything you have run. One word prints it.

Hint 2 (−15 XP): Run `history`.

BOSS: Who Logged the Most Hours? 50 pts · l4-boss · OK

Task: The last thing the day crew wants before they drive up the mountain. `Documents/data.csv` lists each person's hours in column 4. Print the NAME of the person with the most hours — just the name, on its own, with nothing else around it.

Answer: `sort -t, -k4 -nr Documents/data.csv | head -n 1 | cut -d, -f2`

Completion: Auto-completes on: allOf

Hint 1 (free): Three tools on one line: put the rows in order by the hours column with the largest first, keep only the top row, then slice out the name column.

Hint 2 (−15 XP): Run `sort -t, -k4 -nr Documents/data.csv | head -n 1 | cut -d, -f2`.

Also accepted: `sort -t, -k4 -n Documents/data.csv | tail -n 1 | cut -d, -f2`, `sort -t, -k4 -nr Documents/data.csv | head -1 | awk -F, '{print $2}'`

Accepted command variants

Auto-completing challenges accept the natural variations a beginner types, so a correct command is never silently ignored:

- Optional quotes around a path
- Path prefixes: `./Documents`, `~/Documents`, or the full absolute path
- Short paths when already inside the folder
- Flag spacing: `head -n 5`, `head -n5`, `head -5`
- Windows: forward or back slashes, any capitalization (`md5` or `MD5`)
- Trailing spaces are ignored everywhere

Teaching notes

- **The Coach** (toggle in the terminal title bar) explains every command and error in plain language. Explanations fade after the second use of a command. Leave it ON for beginners.
- **Reference tab** holds a manual page for every command, and tells students the real-shell equivalent. Point stuck students there before they buy a hint.
- **Unsimulated tools** reply with what they are, rather than a bare "command not found".
- **Hints:** the first is free and conceptual; the one that gives the exact command costs XP. Hint use is visible in the instructor dashboard as a signal of where the class is struggling.
- **Watch the stuck points.** The dashboard flags any challenge with a solve rate under 35%. Now that briefs no longer contain the answer, that number means something: it marks a concept the class did not get, not a typing error.